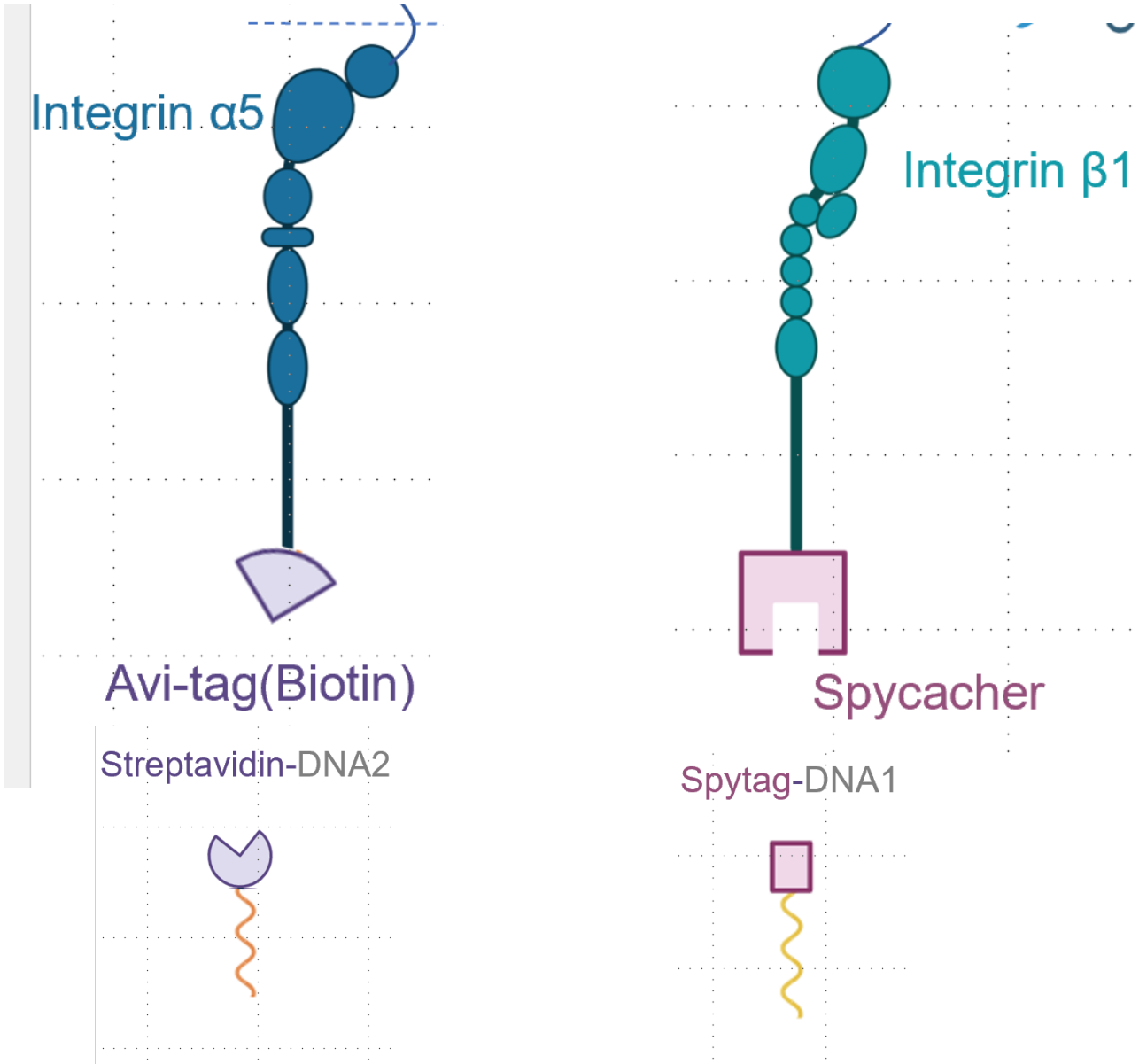
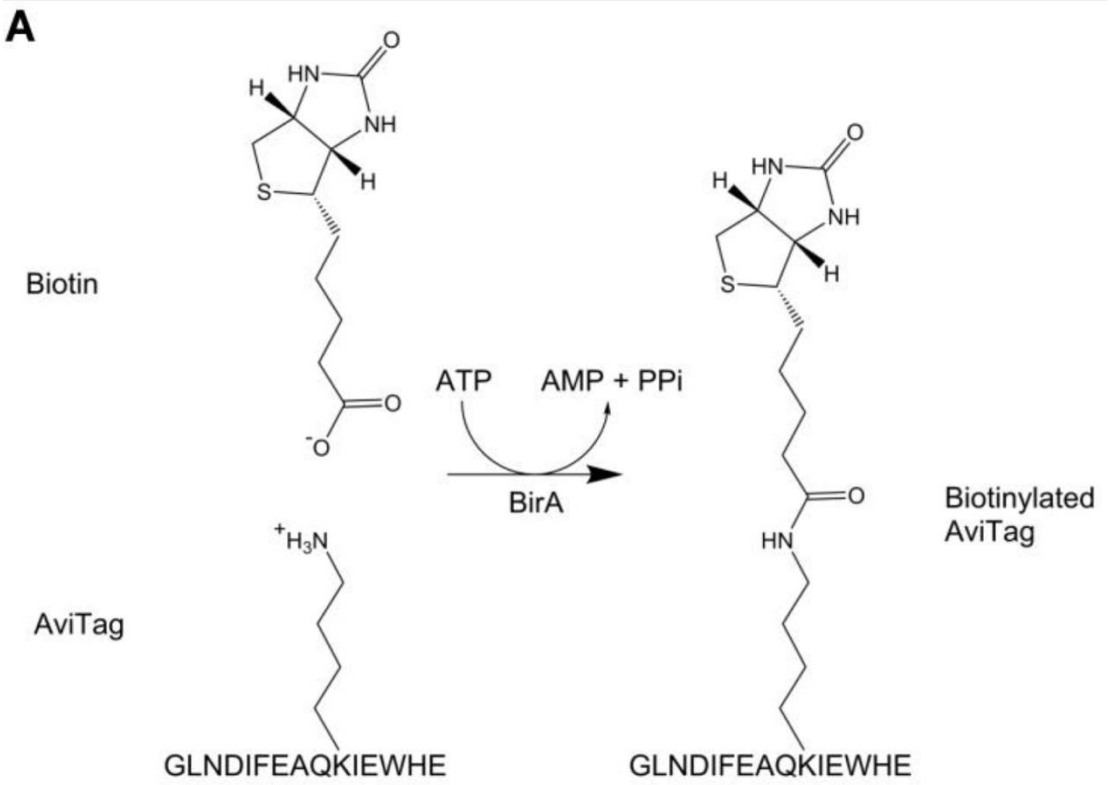
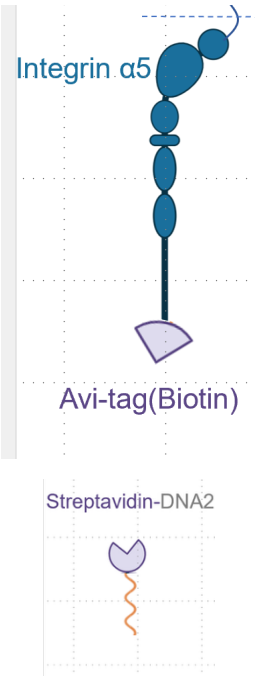
Regarding avb1

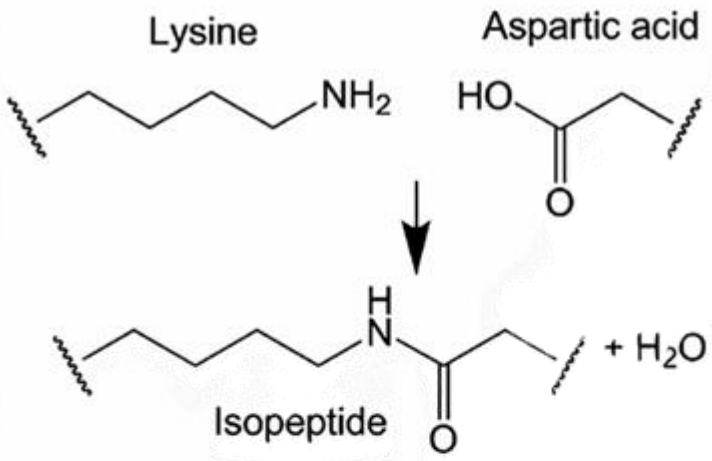




The Avi tag of Integrin α5 -Avi is covalently linked to Biotin, followed by the non-covalent interaction between Biotin and Streptavidin.

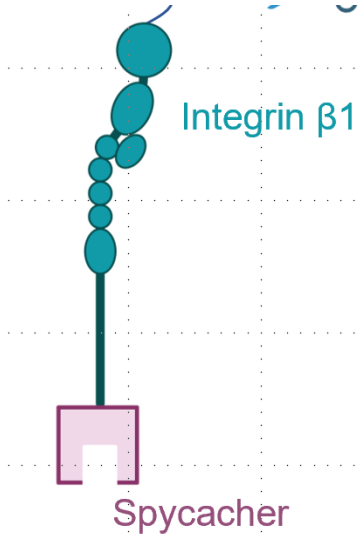
>Integrin α5 -Avi
…GLNDIFEAQKIEWHE

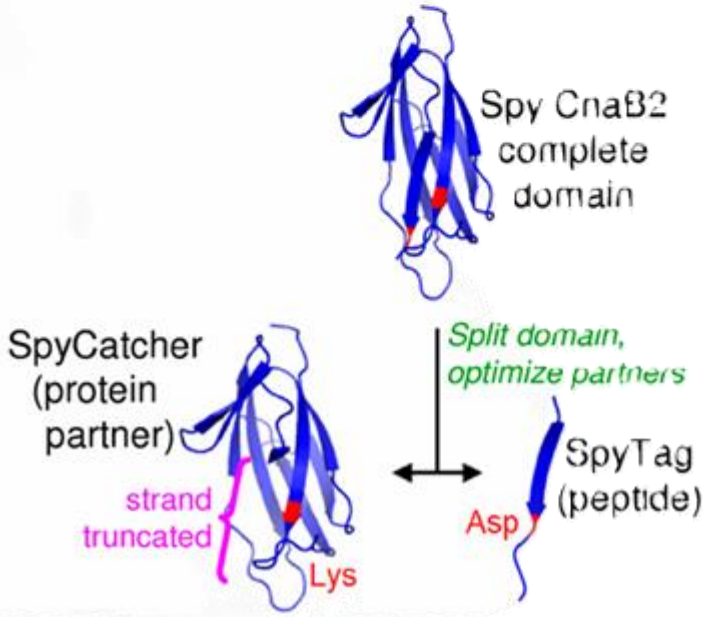




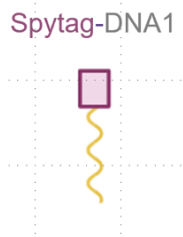
The amino group on the side chain of Lys735 in Integrin β1-SpyCatcher and the carboxyl group on the side chain of Asp10 in SpyTag form an amide bond.

>Integrin β1-spycatcher
…GGGGSGGGGSGGGGSDSATHIKFSKRDEDGKELAG
ATMELRDSSGKTISTWISDGQVKDFYLYPGKYTFVETAA
PDGYEVATAITFTVNEQGQVTVNGKATKGDAHID

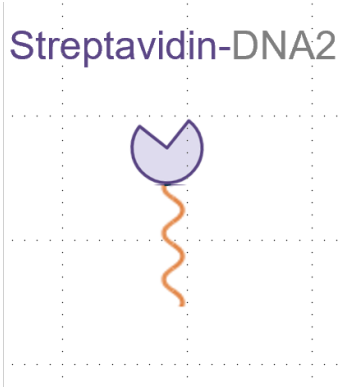


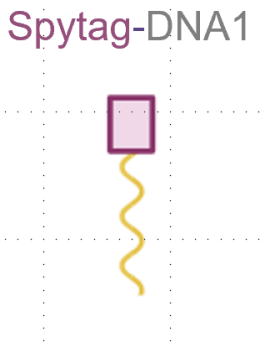


>Spytag
RGVPHIVMVDAYKRYK



Regarding DNA Conjugation



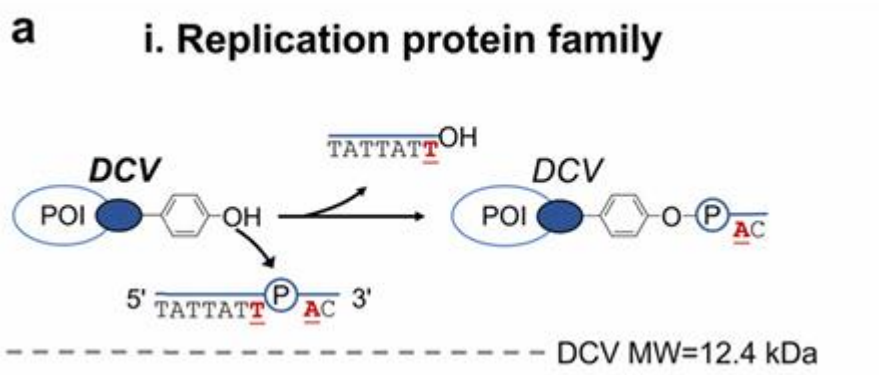


DNA is conjugated to the N-termini of Streptavidin and Spytag.

The attachment can be achieved either via click chemistry or by directly fusing a protein (DCV) to the N-termini of Spytag and Streptavidin. (Please see the next slide for details on DCV).

Our group has not yet finalized which of these two strategies will be used in the current design.

DCV is a protein that recognizes specific single-stranded DNA (ssDNA) sequences and forms a covalent linkage with them.

>dcv
MAKSGNYSYKRWVFTINNPTFE
DYVHVLEFCTLDNCKFAIVGEEK
GANGTPHLQGFLNLRSNARAAA
LEESLGGRAWLSRARGSDEDNE
EYCAKESTYLRVGEPVSKGRSS
DLAEATSAV

While the minimal requirement for DCV recognition is the conserved core nonamer 5'-TAGTATT^AC-3' (where the nicking occurs between the T and A), we generally utilize the 15-mer sequence 5'-AAGTATT^ACCAGAAA-3' as established in the literature.